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Supplementary information

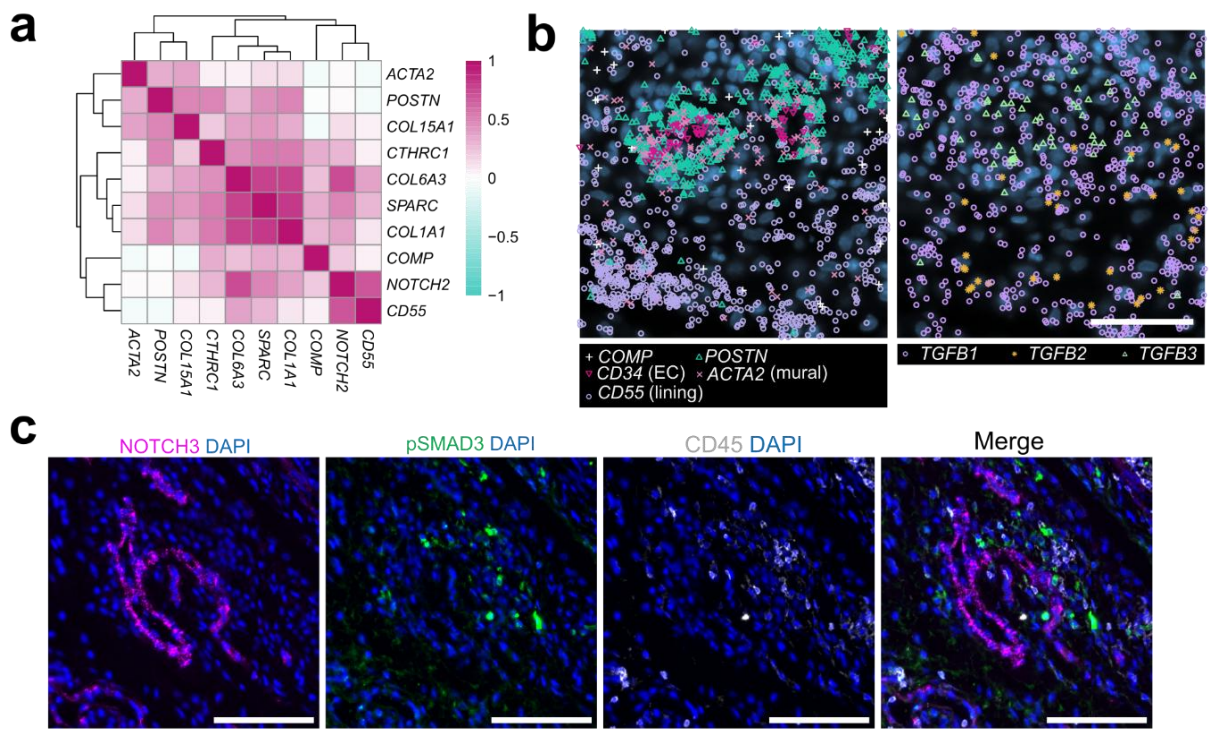
https://doi.org/10.1038/s41590-025-02386-2

Spatial patterning of fibroblast TGFβ signaling underlies treatment resistance in rheumatoid arthritis

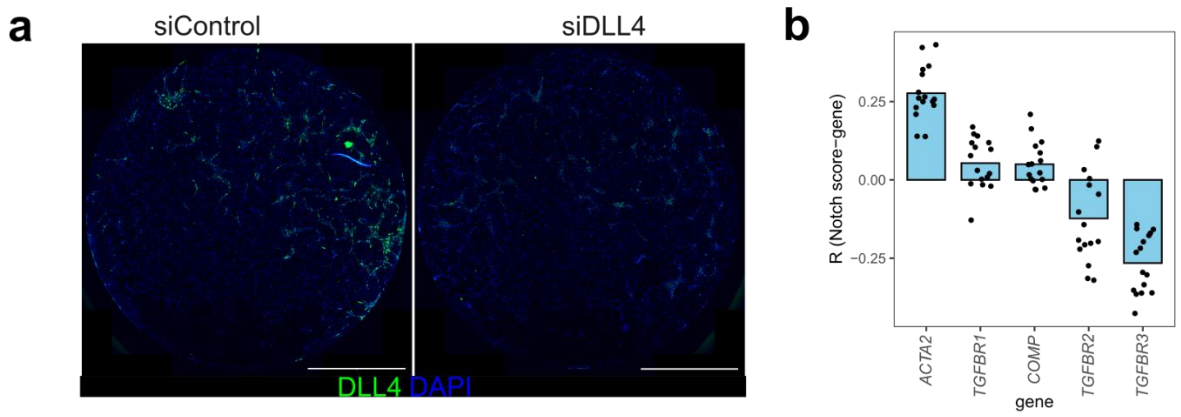
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	Total	remission	non-remission	borderline	remision vs non-remission	
	(N=17)	(N=10)	(N=5)	(N=2)	p-value	significance
Sex						
Female	8 (47%)	4 (40%)	3 (60%)	1 (50%)	0.6084	No
Male	9 (53%)	6 (60%)	2 (40%)	1 (50%)		
Treatment						
DMARD	11 (65%)	7 (70%)	2 (40%)	2 (100%)	0.3287	No
Adalimumab	6 (35%)	3 (30%)	3 (60%)	0 (0%)		
	Mean(SD)	Mean(SD)	Mean(SD)	Mean(SD)		
Age (y)	47.35 (17.45)	50.50 (22.18)	43.20 (3.11)	42.00 (12.73)	0.5941	No
DAS28-ESR (baseline)	5.17 (1.41)	4.89 (1.50)	5.52 (1.04)	5.65 (2.34)	0.4396	No
DAS28-CRP (baseline)	4.69 (1.28)	4.52 (1.34)	5.01 (1.04)	4.77 (2.20)	0.4396	No
TJC53 (baseline)	11.06 (6.68)	9.80 (7.07)	12.60 (3.36)	13.50 (13.44)	0.3570	No
SJC44 (baseline)	9.65 (6.14)	10.10 (7.06)	8.80 (4.71)	9.50 (7.78)	0.9022	No
TJC28 (baseline)	7.24 (5.76)	6.50 (6.65)	8.40 (3.44)	8.00 (8.49)	0.2647	No
SJC28 (baseline)	7.29 (5.52)	7.60 (6.50)	6.80 (4.27)	7.00 (5.66)	1.0000	No
ESR (baseline)	35.59 (25.13)	32.00 (24.81)	30.80 (16.78)	65.50 (37.48)	1.0000	No
PtGA (baseline)	56.18 (28.44)	53.80 (22.66)	65.20 (36.24)	45.50 (48.79)	0.3263	No
VAS Pain (baseline)	53.35 (27.32)	52.10 (25.73)	61.20 (30.08)	40.00 (41.01)	0.5002	No
DAS28-ESR (6 months)	2.59 (1.20)	1.90 (0.50)	4.01 (1.20)	2.49 (0.33)	0.0007	Yes
DAS28-CRP (6 months)	2.41 (1.06)	1.74 (0.37)	3.74 (0.94)	2.45 (0.18)	0.0007	Yes
TJC53 (6 months)	4.82 (9.64)	0.90 (1.91)	14.20 (14.41)	1.00 (0.00)	0.0037	Yes
SJC44 (6 months)	2.24 (2.59)	1.50 (2.12)	3.60 (3.58)	2.50 (0.71)	0.2273	No
TJC28 (6 months)	1.71 (3.41)	0.10 (0.32)	5.20 (4.92)	1.00 (0.00)	0.0011	No
SJC28 (6 months)	2.00 (2.57)	1.30 (1.95)	3.20 (3.83)	2.50 (0.71)	0.4377	No
ESR (6 months)	12.65 (14.33)	8.60 (4.48)	23.20 (23.99)	6.50 (2.12)	0.0964	No
PtGA (6 months)	21.94 (18.36)	15.50 (9.70)	37.80 (26.26)	14.50 (2.12)	0.1107	No
VAS Pain (6 months)	22.00 (18.82)	14.60 (16.21)	39.40 (16.55)	15.50 (7.78)	0.0233	Yes
delta DAS28-ESR	-2.58 (1.56)	-2.99 (1.49)	-1.51 (1.34)	-3.16 (2.02)	0.0400	Yes
delta DAS28-CRP	-2.28 (1.55)	-2.78 (1.45)	-1.27 (1.28)	-2.31 (2.38)	0.0553	No
delta TJC53	-6.24 (10.05)	-8.90 (7.08)	1.60 (11.48)	-12.50 (13.44)	0.1097	No
delta SJC44	-7.41 (6.77)	-8.60 (7.29)	-5.20 (6.46)	-7.00 (7.07)	1.0000	No
delta TJC28	-5.53 (5.94)	-6.40 (6.74)	-3.20 (3.35)	-7.00 (8.49)	0.7108	No
delta SJC28	-5.29 (6.23)	-6.30 (6.91)	-3.60 (5.94)	-4.50 (4.95)	0.9510	No
delta ESR	-22.94 (28.20)	-23.40 (23.49)	-7.60 (26.11)	-59.00 (35.36)	0.4396	No
delta PtGA	-34.24 (28.55)	-38.30 (27.94)	-27.40 (29.23)	-31.00 (46.67)	0.5399	No
delta VAS Pain	-31.35 (34.96)	-37.50 (37.21)	-21.80 (30.78)	-24.50 (48.79)	0.5941	No

Supplementary Table 1. Clinical characteristics of patients in the 396.10 study



Supplementary Figure 1. **a,** Heatmap representing spatial density correlations between fibrogenic and cell type marker genes (mean across 4 samples in the 50-gene custom panel). **b,** Representative *TGFB* isoform transcript expression (right) patterns in 50-gene custom panel (n = 4 RA tissue independently profiled), with cell type and subtype marker genes shown (left) for the same site. Scale bar indicates 50 μm. **c,** Representative immunofluorescence images (n = 3 RA tissue) showing pSMAD3 staining in CD45- cells (i.e. fibroblasts) in a NOTCH3+ perivascular region. Scale bars indicate 200 μm.



Supplementary Figure 2. Validation of DLL4 knockdown and correlation of Notch activation score with vascular niche transcripts. a, Representative RNAscope image showing DLL4 expression in siControl and siDLL4-treated co-culture. Scale bar indicates 2 mm. **b,** Mean two-sided Pearson correlation (R) between Notch activation score (UCell) and transcripts detected per vascular niche for each pre-treatment sample (n = 16). *ACTA2* represents a positive control for expected correlation with Notch signaling. Genes in the Notch activation score include: *HES1, HES4, HEY1, JAG1, NOTCH3, DLL4, NOTCH4, NOTCH1, COL4A1 and COL4A2.*